Patrick Iben, MS

Curriculum vitae

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Research profile

Applied mathematician with 12 years in research, including seven of clinical biostatistics in contract research (nearly four as lead biostatistician on at least 15 early-phase clinical pharmacology studies) and predictive modeling of large-scale healthcare and university data. Works to eliminate needless ambiguity, risk and suffering in decisions that rest on data, by stating in advance how a claim could fail and reporting exactly what was measured and what was not. Develops predictive models, statistical frameworks and inspectable software across clinical biometrics, machine learning, AI evaluation, wearable sensing and public-data monitoring. Open to full-time and consulting roles in research engineering, data science, data architecture and AI/ML.

Professional experience

Biostatistician II | Syneos Health | Remote | December 2021 to October 2026

- Progressed from support statistician to lead biostatistician within a year of joining; led at least 15 early-phase clinical pharmacology studies across first-in-human, single and multiple ascending dose (SAD/MAD), crossover, bioequivalence and bioavailability designs, and supported at least as many more.
- Initiated and created comprehensive SAS validity checks for data coding and derivations at each data stage (ADaM, SDTM and raw) of on-treatment study periods, drawing on clinical resource informatics experience; the checks were very favorably received and adopted by the internal statistical team.
- Owned protocol statistical sections and randomization schedules; contributed to clinical study reports.
- Prepared statistical analysis plans and mock tables, listings and figures; created or reviewed specifications for analysis datasets and outputs using CDISC SDTM/ADaM.
- Performed pharmacokinetic/pharmacodynamic (PK/PD) and non-compartmental analyses (NCA), wrote SAS review programs for quality control and reviewed all tables, listings and figures before delivery.
- Reviewed interim data, annotated case report forms (CRFs) and database designs against analysis plans.

Associate Biostatistician | Emmes | Rockville, MD | September 2019 to December 2021

- Worked on opioid-use-disorder studies for the National Institute on Drug Abuse (NIDA) with nodes of its Clinical Trials Network.
- Created a framework for auto-coding MedDRA and WHODrug codes from manually entered verbatim fields, using natural language processing (NLP) and machine learning.
- Wrote protocol statistical sections, statistical analysis plans and the statistical sections of final study reports and of Data and Safety Monitoring Board (DSMB) reports.
- Built dynamic progress reporting and SAS/R tools for data-management and analysis problems.
- Harmonized electronic health record (EHR) data from four rural clinics for an opioid-use-disorder study and authored the extraction and normalization programs, credited by name in a 2024 *Addiction* paper; supported external research partners.

Research Assistant and Teaching Assistant | University of Colorado Anschutz Medical Campus | Aurora, CO | August 2016 to July 2019

- Constructed, harmonized and maintained multi-state hospital EHR databases with millions of records and hundreds of derived variables as biostatistics research assistant for the Data to Value nursing econometrics initiative at the University of Colorado College of Nursing, to estimate pediatric nursing-care costs; designed the datasets for later reuse.
- Advised about three doctoral nursing candidates on study design and calculated their sample sizes.
- Served for three years as teaching assistant for multiple biostatistics courses, including Biostatistical Methods and Advanced Data Analysis, with office hours, review sessions, occasional lectures and exam proctoring.

Earlier appointments | 2014 to 2016

- **Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers.** Analyses for the Gates Foundation-funded Multi-State Collaborative assessment of student learning outcomes; statistical support on a federal grant proposal.
- **Data Analysis Fellow, Yale School of Child Development.** Evaluated assistive technologies for students with print disabilities.
- **Head Undergraduate Research Assistant, Southern Connecticut State University Office of Assessment and Planning.** Harmonized and analyzed 10 years of university cohort data and developed predictive models of student success and retention; designed student survey questions later validated in statistical modeling as important predictors of academic success.
- **Math Tutor, Southern Connecticut State University.** Tutored undergraduates in statistics, calculus and algebra for a year.

Education and training

- **MS, Biostatistics**, Data Science Analytics minor, University of Colorado Anschutz Medical Campus, 2019. GPA 3.62.
- **BS, Mathematics**, Applications of Mathematics, Southern Connecticut State University, 2016. Magna cum laude; GPA 3.78.
- **High-performance computing training and certification**, University of Colorado Anschutz Medical Campus.
- **Diploma in Audio Production Technologies**, School of Audio Engineering, 2012.

Publications and manuscripts

Submission and review statuses as of 1 November 2026.

Contamination-Resilient Cluster Randomized Trial Design under Spatial Interference: A Simulation Study of Contamination Control. *Statistics in Medicine*, submitted 19 October 2026.

- Developed a design framework for cluster randomized trials in which treatment effects spill between nearby sites, and tested it by simulation on 57 real malaria survey sites in Uganda. Began as master's research at the University of Colorado and resumed years later.
- Preprint doi:10.5281/zenodo.23486176.

Auditing preprocessing-sensitive allele rankings in predicted chromatin contact maps. *BMC Bioinformatics*, submitted 29 October 2026.

- Showed that preprocessing can change allele rankings from an AI genomics model's predicted chromatin contact maps, and built an audit that separates that effect from evidence of better clinical discrimination.
- Integrates an earlier preprint: What do within-site comparisons of AlphaGenome Atlas predictions measure? An audit at ClinVar sites in 11 genes. doi:10.5281/zenodo.23543154.

A false floor: Refuting our own information-limit claim in a lethal identification task. *PLOS ONE*, submitted 22 October 2026.

- Overturned the project's own earlier claim of a hard information limit in a high-stakes identification task, using stronger baselines and checks for overlap with the models' training data.
- Preprint doi:10.5281/zenodo.23519299.

Anticipating Network Contagions: When Directedness Outperforms Connectedness. *Applied Network Science*, with editor.

- Showed that targeting the units that transmit a contagion can beat targeting the hardest-hit ones, using a measure of directional asymmetry across seven contagion datasets (interventions simulated).

Which data modality recognizes a gene set? A domain-agnostic consilience engine. *PeerJ*, in peer review.

- Built an open-source tool that compares how well different kinds of biological data recognize the same gene groups, with statistical testing and replication on nervous-system and immune gene maps.

No emotion-specific image-computable code in the human left amygdala. *Peer Community In*, preprint review.

- Tested whether the human amygdala carries an emotion-specific visual code across three brain-imaging datasets and found no evidence for one, with conclusions limited to the samples and models tested.
- Preprint doi:10.5281/zenodo.21422754.

Published work

- **Acknowledged contributor:** developed data extraction and normalization programs for the study dataset in Osterhage KP, Hser YI, Mooney LJ, Sherman S, Saxon AJ, Ledgerwood M, Holtzer CC, Gehring MA, Clingan SE, Curtis ME, Baldwin LM. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: Challenges and opportunities. *Addiction*. 2024;119(1):160-168. doi:10.1111/add.16338.
- **Co-author:** Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.

Teaching and research communication

- Authored a concise study guide and a complete educational companion for each of the six studies, tied to exact manuscript versions; all are publicly available. The teaching sets include worked examples, answered exercises, glossaries and source-coverage maps.
- Connected manuscripts to code, data deposits, interactive demonstrations and teaching editions through a public research index. The CRT companion spans 97 pages; the others have full methods walkthroughs.
- Developed interactive statistical demonstrations and domain-specific explanations of the ten parameters used in response planning, supporting hover, keyboard focus and touch interaction.

Research engineering and scientific software

Trial biostatistics toolkit and eCRF pipeline apps

A public toolkit for biostatisticians: SAS and R templates, dashboards and quality-control checks for work across an early-phase trial, from the protocol's statistical sections to clinical study report text. It runs from static files or local R, so client data never leave the analyst's machine; the example studies are synthetic.

- Created a complete worked example of a bioequivalence study: one synthetic trial taken from sample size and randomization through the analysis plan, analysis datasets, pharmacokinetic analysis, tables and report text, all regenerated by a single script.
- Designed safety-monitoring tools for first-in-human studies: liver and cardiac screening rules, a Safety Review Committee packet and an interactive dashboard that runs in the browser.
- Engineered a QC harness that recomputes key results a second way and is tested against planted errors.
- Wrote SAS and R program libraries for standard tables, listings and figures across common early-phase designs, plus simulation studies of two regulatory checks.
- Released four Streamlit apps for reviewing electronic case report form (eCRF) data (medical coding, de-identification, participant risk and central monitoring); trained XGBoost models with embedding retrieval for the coding and de-identification paths, evaluated them on labeled synthetic data and published their measured limitations.

Public Data Consoles

- Built 36 self-contained, browser-only decision-support consoles spanning hazards, infrastructure, health-system strain, public health, access to care and markets; each labels every input as measured, modeled or demonstration. Examples include river-flood propagation, earthquake consequences, U.S. grid strain, NHS Scotland hospital strain, CDC surveillance and the World Bank global-health atlas.
- Integrated public observations, source-issued forecasts, published estimates and dated snapshots; implemented scheduled data publication, schema and timestamp checks, fallback behavior and source-specific freshness displays. Visitors need no provider API key.
- Adapted the ten-parameter planning framework across all 36 consoles. Response plans prioritize supported actions using the selected data, with review timing, escalation triggers and public/operator tracks.
- Added mobile and keyboard/touch-accessible explanations, with catalog, unit and browser checks.

Coastal Water Resilience

- Unified two original research methods in Coastal Water Resilience: the ten-metric diagnostic suite from a contamination-resilient cluster randomized trial design, combined with directed source/sink topology accounting, applied to live NOAA, NHC and NWS coastal forecasts across 131 published station locations, with terrain, evacuation zones and Census context driving projected impacts and response plans.
- Joined official public wastewater-station inventories to evacuation zones, USGS terrain and source-defined sewer basins, alongside NOAA/NWS observations and forecasts. Linked projected-impact screens to the same station scope used by the ten-parameter response plans.

- Incorporated Census population context and published annual treatment flows while distinguishing geographic exposure from operational outages, customer losses and hydraulic connectivity. The synthetic exercise remains a separately labeled demonstration.
- Connected parameters to phase-specific actions, explaining why a diagnostic matters at a particular timepoint and when to recheck or escalate.

Autonomic Telemetry

- Prototyped a personal physiological-arousal biofeedback system for work and rest: a multivariate arousal estimate combining heart rate, heart-rate variability, EEG beta/alpha and covariance distance against a two-minute personal reference, with prediction-error event detection; a local Mac-to-iPhone bridge streams live input from a Muse S headband and a Hume Band, and a Signal Lab replays recordings.
- Implemented source identity, freshness, signal-quality and channel-coverage controls, plus missing-channel handling and replayable feature exports. The physiological-pattern estimate is experimental.
- Built a separate Signal Lab with numerical lenses, analytic checks and regression tests.

Resilience Telemetry

- Applied original directed-contagion research to cross-asset market volatility: spillover networks, transmitter stability and historical replay, with held-out tests of predictive value against baselines that track the loudest and most persistent assets.
- Integrated public market and macroeconomic data with source timestamps and interpretation limits. The ridge-regularized VAR/FEVD implementation is an exploratory adaptation of the network-contagion research; the manuscript's non-negative VAR intervention experiments are documented separately.

Consilience Engine

- Packaged gene-set analysis as an installable Python library and configurable command-line tool with offline cache support. Constructed modality-specific graphs from InterPro, Human Protein Atlas, STRING and Gene Ontology sources, with permutation-null separation and class-recovery analyses.
- Applied the framework to a 410-gene nervous-system map and a 132-gene immune replication, with unit, property and metamorphic tests and clustering-sensitivity analyses.

catmech R package

- Submitted catmech to CRAN in November 2026, an R package for regression when only a subset of records has been checked against a reference.

Programming and evaluation skills

- **Programming and engineering:** Python (PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, pandas, statsmodels, FastAPI, Streamlit), R (tidyverse, pharmaverse and Shiny), Quarto, SAS (Base, Macro, STAT, SQL), SQL, JavaScript, LaTeX, Git, Docker, high-performance computing, public API integration and browser testing.
- **Data science and AI/ML:** predictive analytics, machine learning, natural language processing, data architecture and harmonization, real-world data and decision-support dashboards.
- **AI evaluation:** numeric grounding, provenance auditing, retrieval evaluation, model comparison, synthetic-oracle checks and reproducibility testing.

Clinical expertise

- **Therapeutic areas:** cardiovascular, dermatology, immunology, neurology (MS, ALS, Parkinson's, Alzheimer's), oncology (hematologic malignancies and solid tumors), psychiatry (opioid use disorder and schizophrenia), and respiratory.
- **Study designs:** first-in-human, SAD/MAD, crossover, fixed sequence, parallel group, adaptive, placebo-controlled, double-blind, open-label and cohort.
- **Statistical methods:** PK/PD and NCA, bioequivalence, survival and longitudinal analysis, missing-data imputation, simulation, permutation and resampling inference, multiplicity, cluster randomized designs, causal inference under interference (spillover and contamination, directed network interventions), consilience testing across independent data modalities, calibration and decision curves.
- **Clinical standards and tools:** CDISC SDTM/ADaM, MedDRA and WHODrug coding, Veeva Vault and TLF review.